Complex Trigonometric Functions

If x is a real variable, then

$$e^{ix} = \cos x + i \sin x \quad \text{and} \quad e^{-ix} = \cos x - i \sin x. \tag{1}$$

By adding these equations and simplifying, we obtain an equation that relates the real cosine function with the complex exponential function:

$$\cos x = \frac{e^{ix} + e^{-ix}}{2}. \tag{2}$$

In a similar manner, if we subtract the two equations in (1), then we obtain an expression for the real sine function:

$$\sin x = \frac{e^{ix} - e^{-ix}}{2i}. \tag{3}$$

The formulas for the *real* cosine and sine functions given in (2) and (3) can be used to define the *complex* sine and cosine functions. Namely, we define these complex trigonometric functions by replacing the real variable x with the complex variable z in (2) and (3). This discussion is summarized in the following definition.

Definition [redacted] Complex Sine and Cosine Functions

The complex sine and cosine functions are defined by:

$$\sin z = \frac{e^{iz} - e^{-iz}}{2i} \quad \text{and} \quad \cos z = \frac{e^{iz} + e^{-iz}}{2}. \tag{4}$$

($e^{iz} = \cos z + i.\sin z$, $e^{-iz} = \cos z - i.\sin z$)

It follows from (2) and (3) that the complex sine and cosine functions defined by (4) agree with the real sine and cosine functions for real input. Analogous to real trigonometric functions, we next define the complex tangent, cotangent, secant, and cosecant functions using the complex sine and cosine:

$$\tan z = \frac{\sin z}{\cos z}, \quad \cot z = \frac{\cos z}{\sin z}, \quad \sec z = \frac{1}{\cos z}, \quad \text{and} \quad \csc z = \frac{1}{\sin z}. \tag{5}$$

These functions also agree with their real counterparts for real input.

EXAMPLE 1 **Values of Complex Trigonometric Functions**

In each part, express the value of the given trigonometric function in the form $a + ib$.

(a) $\cos i$ **(b)** $\sin(2+i)$ **(c)** $\tan(\pi - 2i)$

Solution For each expression we apply the appropriate formula from (4) or (5) and simplify.

(a) By (4),

$$\cos i = \frac{e^{i \cdot i} + e^{-i \cdot i}}{2} = \frac{e^{-1} + e}{2} \approx 1.5431.$$

(b) By (4),

$$\begin{aligned} \sin(2+i) &= \frac{e^{i(2+i)} - e^{-i(2+i)}}{2i} \\ &= \frac{e^{-1+2i} - e^{1-2i}}{2i} \\ &= \frac{e^{-1}(\cos 2 + i \sin 2) - e(\cos(-2) + i \sin(-2))}{2i} \\ &\approx \frac{0.9781 + 2.8062i}{2i} \\ &\approx 1.4031 - 0.4891i. \end{aligned}$$

(**c**) By the first entry in (5) together with (4) we have:

$$\tan(\pi - 2i) = \frac{\left(e^{i(\pi-2i)} - e^{-i(\pi-2i)}\right)/2i}{\left(e^{i(\pi-2i)} + e^{-i(\pi-2i)}\right)/2} = \frac{e^{i(\pi-2i)} - e^{-i(\pi-2i)}}{\left(e^{i(\pi-2i)} + e^{-i(\pi-2i)}\right) i}$$

$$= -\frac{e^2 - e^{-2}}{e^2 + e^{-2}} i \approx -0.9640i.$$

Identities : Trigonometriden bildiğimiz özdesliklerin aynısı

$$\sin(-z) = -\sin z \qquad \cos(-z) = \cos z \tag{6}$$

$$\cos^2 z + \sin^2 z = 1 \tag{7}$$

$$\sin(z_1 \pm z_2) = \sin z_1 \cos z_2 \pm \cos z_1 \sin z_2 \tag{8}$$

$$\cos(z_1 \pm z_2) = \cos z_1 \cos z_2 \mp \sin z_1 \sin z_2 \tag{9}$$

Observe that the double-angle formulas:

$$\sin 2z = 2\sin z \cos z \qquad \cos 2z = \cos^2 z - \sin^2 z \tag{10}$$

follow directly from (8) and (9).

We will verify only identity (7). The other identities follow in a similar manner.

$$\cos^2 z = \left(\frac{e^{iz} + e^{-iz}}{2}\right)^2 = \frac{e^{2iz} + 2 + e^{-2iz}}{4},$$

and

$$\sin^2 z = \left(\frac{e^{iz} - e^{-iz}}{2i}\right)^2 = -\frac{e^{2iz} - 2 + e^{-2iz}}{4}.$$

$$\cos^2 z + \sin^2 z = \frac{e^{2iz} + 2 + e^{-2iz} - e^{2iz} + 2 - e^{-2iz}}{4} = 1.$$

It is important to recognize that some properties of the real trigonometric functions are *not* satisfied by their complex counterparts. For example, $|\sin x| \leq 1$ and $|\cos x| \leq 1$ for all real x, but, from Example 1 we have $|\cos i| > 1$ and $|\sin(2+i)| > 1$ since $|\cos i| \approx 1.5431$ and $|\sin(2+i)| \approx 1.4859$, so these inequalities, in general, are not satisfied for complex input.

Sin ve cos fonksiyonları kompleks input için ≤ 1 olmak zorunda değil!

Ex: $|\cos i| \approx 1.5431 > 1$

Periodicity we proved that the complex exponential function is periodic with a pure imaginary period of $2\pi i$. That is, we showed that $e^{z+2\pi i} = e^z$ for all complex z. Replacing z with iz in this equation we obtain $e^{iz+2\pi i} = e^{i(z+2\pi)} = e^{iz}$. Thus, e^{iz} is a periodic function with real period 2π. Similarly, we can show that $e^{-i(z+2\pi)} = e^{-iz}$ and so e^{-iz} is also a periodic function with a real period of 2π.

$$\sin(z+2\pi) = \frac{e^{i(z+2\pi)} - e^{-i(z+2\pi)}}{2i} = \frac{e^{iz} - e^{-iz}}{2i} = \sin z.$$

A similar statement also holds for the complex cosine function. In summary, we have:

$$\sin(z+2\pi) = \sin z \quad \text{and} \quad \cos(z+2\pi) = \cos z \tag{11}$$

Periods: e^z: $2\pi i$ (Pure Imaginery), e^{iz}: 2π, $\sin z$: 2π → Real Periods

$$\sin(z + 2\pi) = \sin z \quad \text{and} \quad \cos(z + 2\pi) = \cos z \qquad (11)$$

for all z. Put another way, (11) shows that the complex sine and cosine are periodic functions with a real period of 2π. The periodicity of the secant and cosecant functions follows immediately from (11) and (5). The identities $\sin(z + \pi) = -\sin z$ and $\cos(z + \pi) = -\cos z$ can be used to show that the complex tangent and cotangent are periodic with a real period of π.

Trigonometric Equations We now turn our attention to solving simple trigonometric equations. Because the complex sine and cosine functions are periodic, there are always infinitely many solutions to equations of the form $\sin z = w$ or $\cos z = w$.

EXAMPLE 2 Solving Trigonometric Equations

Find all solutions to the equation $\sin z = 5$.

Solution By Definition the equation $\sin z = 5$ is equivalent to the equation

$$\frac{e^{iz} - e^{-iz}}{2i} = 5. \Rightarrow \frac{e^{iz} - \frac{1}{e^{iz}}}{2i} \Rightarrow \frac{e^{2iz} - 1}{e^{iz} \cdot 2i} = 5$$

By multiplying this equation by e^{iz} and simplifying we obtain

$$e^{2iz} - 10ie^{iz} - 1 = 0.$$

This equation is quadratic in e^{iz}. That is,

$$e^{2iz} - 10ie^{iz} - 1 = \left(e^{iz}\right)^2 - 10i\left(e^{iz}\right) - 1 = 0.$$

Thus, it follows from the quadratic formula that the solutions of $e^{2iz} - 10ie^{iz} - 1 = 0$ are given by

$b^2 - 4ac$

$$e^{iz} = \frac{10i + (-96)^{1/2}}{2} = 5i \pm 2\sqrt{6}i = \left(5 \pm 2\sqrt{6}\right)i. \tag{12}$$

In order to find the values of z satisfying (12), we solve the two exponential equations in (12) using the complex logarithm. If $e^{iz} = (5 + 2\sqrt{6})\,i$, then $iz = \ln\left(5i + 2\sqrt{6}i\right)$ or $z = -i\ln\left[(5 + 2\sqrt{6})\,i\right]$. Because $(5 + 2\sqrt{6})\,i$ is a pure imaginary number and $5 + 2\sqrt{6} > 0$, we have $\arg\left[(5 + 2\sqrt{6})\,i\right] = \frac{1}{2}\pi + 2n\pi$. Thus,

$$z = -i\log\left[\left(5 + 2\sqrt{6}\right)i\right] = -i\left[\log_e\left(5 + 2\sqrt{6}\right) + i\left(\frac{\pi}{2} + 2n\pi\right)\right]$$

or

$$z = \frac{(4n+1)\pi}{2} - i\log_e\left(5 + 2\sqrt{6}\right) \tag{13}$$

for $n = 0, \pm 1, \pm 2, \ldots$. In a similar manner, we find that if $e^{iz} = (5 - 2\sqrt{6})\,i$, then $z = -i\ln\left[(5 - 2\sqrt{6})\,i\right]$. Since $(5 - 2\sqrt{6})\,i$ is a pure imaginary number and $5 - 2\sqrt{6} > 0$, it has an argument of $\pi/2$, and so:

$$z = -i \log\left[\left(5 - 2\sqrt{6}\right) i\right] = -i\left[\log_e\left(5 - 2\sqrt{6}\right) + i\left(\frac{\pi}{2} + 2n\pi\right)\right]$$

or

$$z = \frac{(4n+1)\pi}{2} - i \log_e\left(5 - 2\sqrt{6}\right) \tag{14}$$

for $n = 0, \pm1, \pm2, \ldots$. Therefore, we have shown that if $\sin z = 5$, then z is one of the values given in (13) or (14).

Modulus The modulus of a complex trigonometric function can also be helpful in solving trigonometric equations. To find a formula in terms of x and y for the modulus of the sine and cosine functions, we first express these functions in terms of their real and imaginary parts. If we replace the symbol z with the symbol $x + iy$ in the expression for $\sin z$ in (4), then we obtain:

$$\begin{aligned} \sin z &= \frac{e^{-y+ix} - e^{y-ix}}{2i} = \frac{e^{-y}(\cos x + i\sin x) - e^{y}(\cos x - i\sin x)}{2i} \\ &= \sin x\left(\frac{e^{y}+e^{-y}}{2}\right) + i\cos x\left(\frac{e^{y}-e^{-y}}{2}\right). \end{aligned} \tag{15}$$

Since the real hyperbolic sine and cosine functions are defined by $\sinh y = \dfrac{e^{y}-e^{-y}}{2}$ and $\cosh y = \dfrac{e^{y}+e^{-y}}{2}$ we can rewrite (15) as

$$\sin z = \sin x \cosh y + i \cos x \sinh y. \tag{16}$$

A similar computation enables us to express the complex cosine function in terms of its real and imaginary parts as:

$$\cos z = \cos x \cosh y - i \sin x \sinh y. \tag{17}$$

We now use (16) and (17) to derive formulas for the modulus of the complex sine and cosine functions. From (16) we have:

$$|\sin z| = \sqrt{\sin^2 x \cosh^2 y + \cos^2 x \sinh^2 y}.$$

This formula can be simplified using the identities $\cos^2 x + \sin^2 x = 1$ and $\cosh^2 y = 1 + \sinh^2 y$ for the real trigonometric and hyperbolic functions:

$$
\begin{aligned}
|\sin z| &= \sqrt{\sin^2 x \left(1 + \sinh^2 y\right) + \cos^2 x \sinh^2 y} \\
&= \sqrt{\sin^2 x + \left(\cos^2 x + \sin^2 x\right) \sinh^2 y},
\end{aligned}
$$

or

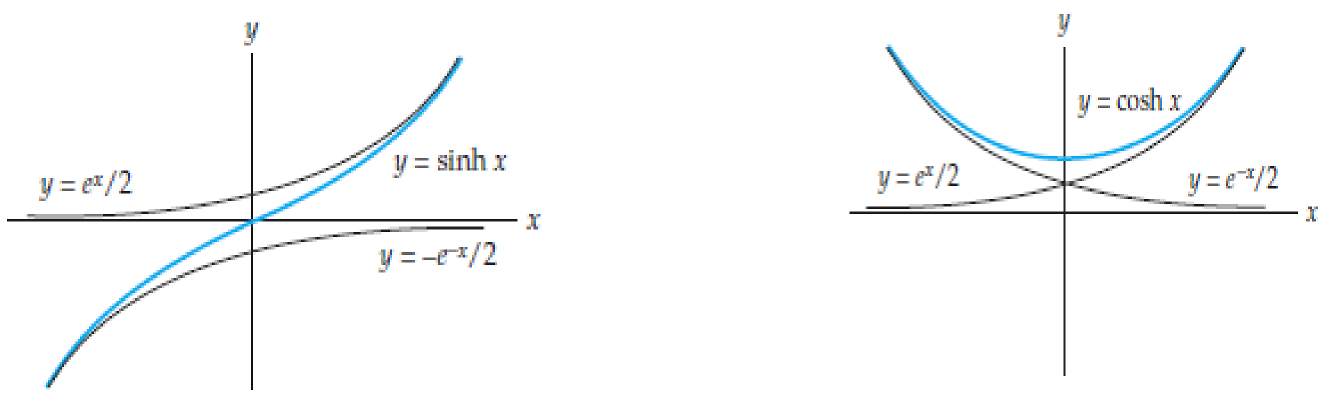
$$
|\sin z| = \sqrt{\sin^2 x + \sinh^2 y}.
$$

Kompleks değişkenli sin ve cos için ≤1 diyemiyor olmamızın sebebi bu

After a similar computation we obtain the following expression for the modulus of the complex cosine function:

$$
|\cos z| = \sqrt{\cos^2 x + \sinh^2 y}. \tag{19}
$$

You may recall from calculus that the real hyperbolic function $\sinh x$ is unbounded on the real line. See Figure 1(a). As a result of this fact, the expressions in (18) and (19) can be made arbitrarily large by choosing y to be arbitrarily large. Thus, we have shown that the complex sine and cosine functions are not bounded on the complex plane. That is, there does not exist a real constant M so that $|\sin z| < M$ for all z in $\mathbf{C}$, nor does there exist a real constant M so that $|\cos z| < M$ for all z in $\mathbf{C}$. This, of course, is quite different from the situation for the real sine and cosine functions for which $|\sin x| \leq 1$ and $|\cos x| \leq 1$ for all real x.

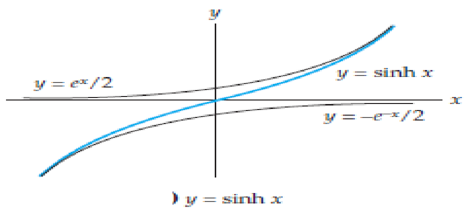


(a) $y = \sinh x$

(b) $y = \cosh x$

Zeros

Using (18) we see that if $|\sin z| = 0$, then $\sqrt{\sin^2 x + \sinh^2 y} = 0$, which is equivalent to:



$$\sin^2 x + \sinh^2 y = 0.$$

Since $\sin^2 x$ and $\sinh^2 y$ are both nonnegative real numbers, this last equation is satisfied if and only if $\sin x = 0$ and $\sinh y = 0$. As just noted, $\sin x = 0$ when $x = n\pi$, $n = 0, \pm 1, \pm 2, \ldots$, and inspection of Figure indicates that $\sinh y = 0$ only when $y = 0$. Therefore, the only solutions of the equation $\sin z = 0$ in the complex plane are the real numbers $z = n\pi$, $n = 0, \pm 1, \pm 2, \ldots$. That is, the zeros of the complex sine function are the same as the zeros of the real sine functions; there are no additional zeros of sine in the complex plane. This is not the same as the situation for polynomial functions where there are often additional zeros in the complex plane.

In essentially the same manner we can show that the only zeros of the complex cosine function are the real numbers $z = (2n+1)\pi/2$, $n = 0, \pm 1, \pm 2 \ldots$.

Thus,

$$\sin z = 0 \text{ if and only if } z = n\pi, \tag{20}$$

and

$$\cos z = 0 \text{ if and only if } z = \frac{(2n+1)\pi}{2} \tag{21}$$

for $n = 0, \pm 1, \pm 2, \ldots$.

Analyticity The derivatives of the complex sine and cosine functions are found using the chain rule

$$\frac{d}{dz}\sin z = \frac{d}{dz}\left(\frac{e^{iz} - e^{-iz}}{2i}\right) = \frac{ie^{iz} + ie^{-iz}}{2i} = \frac{e^{iz} + e^{-iz}}{2},$$

or

$$\frac{d}{dz}\sin z = \cos z.$$

Since this derivative is defined for all complex z, $\sin z$ is an entire function. In a similar manner, we find

$$\frac{d}{dz}\cos z = -\sin z.$$

The derivatives of $\sin z$ and $\cos z$ can then be used to show that derivatives of all of the complex trigonometric functions are the same as derivatives of the real trigonometric functions. The derivatives of the six complex trigonometric functions are summarized in the following.

Tek fark x yerine z var.

Derivatives of Complex Trigonometric Functions

$$\frac{d}{dz}\sin z = \cos z \qquad \frac{d}{dz}\cos z = -\sin z$$

$$\frac{d}{dz}\tan z = \sec^2 z \qquad \frac{d}{dz}\cot z = -\csc^2 z$$

$$\frac{d}{dz}\sec z = \sec z \tan z \qquad \frac{d}{dz}\csc z = -\csc z \cot z$$

The sine and cosine functions are entire, but the tangent, cotangent, secant, and cosecant functions are only analytic at those points where the denominator is nonzero. From (20) and (21), it then follows that the tangent and secant functions have singularities at $z = (2n+1)\pi/2$ for $n = 0, \pm 1, \pm 2 \ldots$, whereas the cotangent and cosecant functions have singularities at $z = n\pi$ for $n = 0, \pm 1, \pm 2 \ldots$.

Trigonometric Mapping

Describe the image of the region $-\pi/2 \le x \le \pi/2$, $-\infty < y < \infty$, under the complex mapping $w = \sin z$.

Solution one approach to this problem is to determine the image of vertical lines $x = a$ with $-\pi/2 \le a \le \pi/2$ under $w = \sin z$. Assume for the moment that $a \ne -\pi/2$, 0, or $\pi/2$. From (16) the image of the vertical line $x = a$ under $w = \sin z$ is given by:

$$u = \sin a \cosh y, \quad v = \cos a \sinh y, \quad -\infty < y < \infty. \tag{22}$$

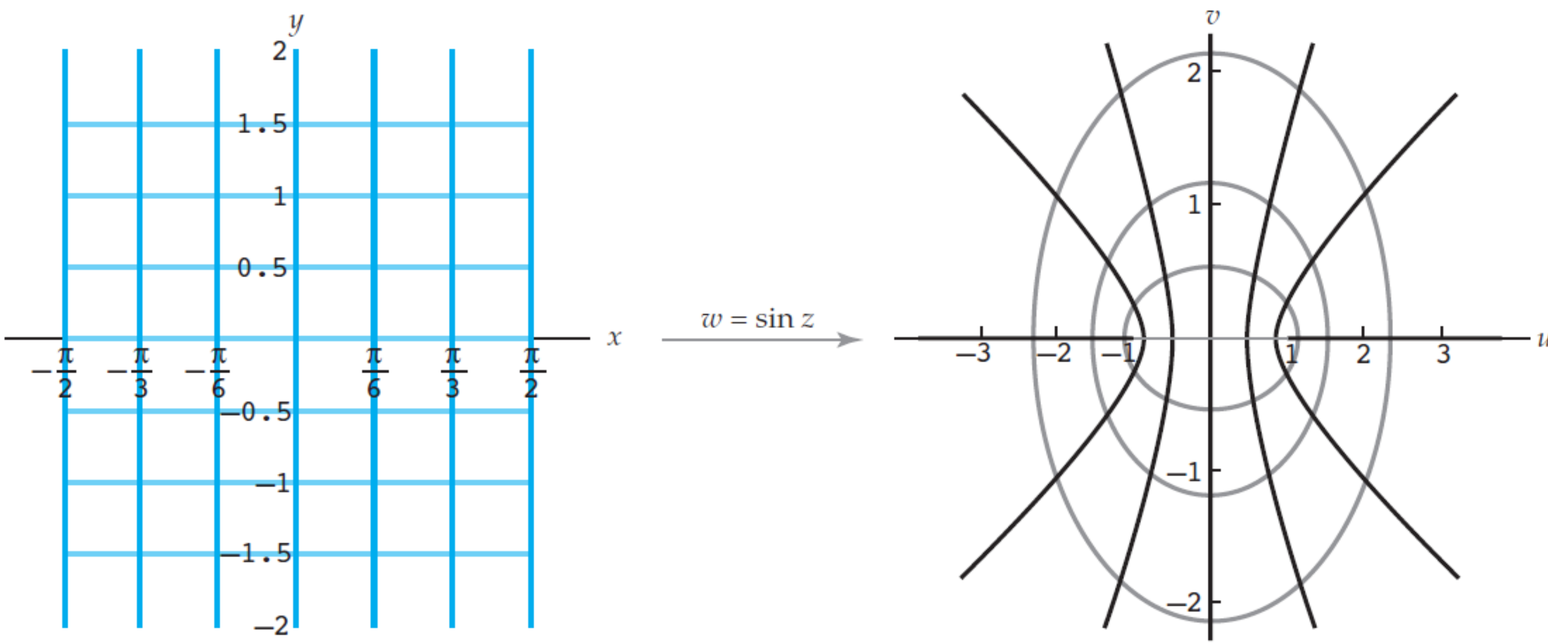
We will eliminate the variable y in (22) to obtain a single Cartesian equation relating u and v. Since $-\pi/2 < a < \pi/2$ and $a \ne 0$, it follows that $\sin a \ne 0$ and $\cos a \ne 0$, and so from (22) we obtain $\cosh y = \dfrac{u}{\sin a}$ and $\sinh y = \dfrac{v}{\cos a}$. The identity $\cosh^2 y - \sinh^2 y = 1$ for real hyperbolic functions then gives the following equation:

$$\left(\frac{u}{\sin a}\right)^2 - \left(\frac{v}{\cos a}\right)^2 = 1. \tag{23}$$

The Cartesian equation in (23) is a hyperbola with vertices at $(\pm \sin a,\ 0)$ and slant asymptotes $v = \pm\left(\dfrac{\cos a}{\sin a}\right)u$. Because the point $(a,\ 0)$ is on the line $x = a$, the point $(\sin a,\ 0)$ must be on the image of the line. Therefore, the image of the vertical line $x = a$ with $-\pi/2 < a < \pi/2$ and $a \neq 0$ under $w = \sin z$ is the branch[†] of the hyperbola (23) that contains the point $(\sin a, 0)$. Because $\sin(-z) = -\sin z$ for all z, it also follows that the image of the line $x = -a$ is branch of the hyperbola (23) containing the point $(-\sin a,\ 0)$.

Therefore, the pair of vertical lines $x = a$ and $x = -a$, with $-\pi/2 < a < \pi/2$ and $a \neq 0$, are mapped onto the hyperbola given by (23). We illustrate this mapping property of $w = \sin z$ in Figure , where the vertical lines shown in color in Figure (a) are mapped onto the hyperbolas shown in black in Figure (b). The line $x = \pi/3$ is mapped onto the branch of the hyperbola containing the point $\left(\frac{1}{2}\sqrt{3}, 0\right)$ and the line $x = \pi/6$ is mapped onto the branch containing the point $\left(\frac{1}{2},\ 0\right)$. Similarly, the line $x = -\pi/3$ is mapped onto the branch containing the point $\left(-\frac{1}{2}\sqrt{3},\ 0\right)$ and the line $x = -\pi/6$ is mapped onto the branch containing the point $\left(-\frac{1}{2},\ 0\right)$.

The images of the lines $x = -\pi/2$, $x = \pi/2$ and $x = 0$ cannot be found from (23). However, from (22) we see that the image of the line $x = -\pi/2$ is the set of points $u \leq -1$ on the negative real axis, that the image of the line $x = \pi/2$ is the set of points $u \geq 1$ on the positive real axis, and that the image of the line $x = 0$ is the imaginary axis $u = 0$. See Figure In summary, we have shown that the image of the infinite vertical strip $-\pi/2 \leq x \leq \pi/2$, $-\infty < y < \infty$, under $w = \sin z$, is the entire w-plane.



(a) The region $-\pi/2 \leq x \leq \pi/2$

(b) Image of the region in (a)

the image could also be found using horizontal line segments $y = b$, $-\pi/2 \leq x \leq \pi/2$, instead of vertical lines. In this case, the images are given by:

$$u = \sin x \cosh b, \quad v = \cos x \sinh b, \quad -\frac{\pi}{2} < x < \frac{\pi}{2}.$$

When $b \neq 0$, this set is also given by the Cartesian equation:

$$\left(\frac{u}{\cosh b}\right)^2 + \left(\frac{v}{\sinh b}\right)^2 = 1, \tag{24}$$

which is an ellipse with u-intercepts at $(\pm \cosh b,\ 0)$ and v-intercepts at $(0, \pm \sinh b)$. If $b > 0$, then the image of the line segment $y = b$ is the upper-half of the ellipse defined by (24) and the image of the line segment $y = -b$ is the bottom-half of the ellipse. Thus, the horizontal line segments shown in color in Figure (a) are mapped onto the ellipses shown in black in Figure (b). The innermost pair of horizontal line segments are mapped onto the innermost ellipse, the middle pair of line segments are mapped onto the middle ellipse, and the outermost pair of line segments are mapped onto the outermost ellipse. As a final point, observe that if $b = 0$, then the image of the line segment $y = 0$, $-\pi/2 < x < \pi/2$, is the line segment $-1 \leq u \leq 1$, $v = 0$ on the real axis.

Complex Hyperbolic Functions

The *real* hyperbolic sine and hyperbolic cosine functions are defined using the *real* exponential function as follows:

$$\sinh x = \frac{e^x - e^{-x}}{2} \quad \text{and} \quad \cosh x = \frac{e^x + e^{-x}}{2}.$$

The *complex* hyperbolic sine and cosine functions are defined in an analogous manner using the *complex* exponential function.

Definition [redacted] Complex Hyperbolic Sine and Cosine

The complex **hyperbolic sine** and **hyperbolic cosine** functions are defined by:

$$\sinh z = \frac{e^z - e^{-z}}{2} \quad \text{and} \quad \cosh z = \frac{e^z + e^{-z}}{2}. \tag{25}$$

Since the complex exponential function agrees with the real exponential function for real input, it follows from (25) that the complex hyperbolic functions agree with the real hyperbolic functions for real input.

The complex hyperbolic tangent, cotangent, secant, and cosecant are defined in terms of $\sinh z$ and $\cosh z$:

$$\tanh z = \frac{\sinh z}{\cosh z}, \quad \coth z = \frac{\cosh z}{\sinh z}, \quad \operatorname{sech} z = \frac{1}{\cosh z}, \quad \text{and } \operatorname{csch} z = \frac{1}{\sinh z}. \tag{26}$$

Observe that the hyperbolic sine and cosine functions are entire because the functions e^z and e^{-z} are entire. Moreover, from the chain rule , we have:

$$\frac{d}{dz}\sinh z = \frac{d}{dz}\left(\frac{e^z - e^{-z}}{2}\right) = \frac{e^z + e^{-z}}{2}$$

or

$$\frac{d}{dz}\sinh z = \cosh z.$$

A similar computation for $\cosh z$ yields

$$\frac{d}{dz}\cosh z = \sinh z.$$

Derivatives of Complex Hyperbolic Functions

$$\frac{d}{dz}\sinh z = \cosh z \qquad \frac{d}{dz}\cosh z = \sinh z$$

$$\frac{d}{dz}\tanh z = \operatorname{sech}^2 z \qquad \frac{d}{dz}\coth z = -\operatorname{csch}^2 z$$

$$\frac{d}{dz}\operatorname{sech} z = -\operatorname{sech} z \tanh z \qquad \frac{d}{dz}\operatorname{csch} z = -\operatorname{csch} z \coth z$$

Relation To Sine and Cosine

The real trigonometric and the real hyperbolic functions share many similar properties. For example,

$$\frac{d}{dx}\sin x = \cos x \quad \text{and} \quad \frac{d}{dx}\sinh x = \cosh x.$$

Aside from the similar notational appearance and the similarities of their respective Taylor series, there is no simple way to relate the real trigonometric and the real hyperbolic functions. When dealing with the *complex* trigonometric and hyperbolic functions, however, there is a simple and beautiful connection between the two. We derive this relationship by replacing z with iz in the definition of $\sinh z$ and then comparing the result with (4):

$$\sinh(iz) = \frac{e^{iz} - e^{-iz}}{2} = i\left(\frac{e^{iz} - e^{-iz}}{2i}\right) = i\sin z,$$

or

$$-i\sinh(iz) = \sin z.$$

In a similar manner, if we substitute iz for z in the expression for $\sin z$ and compare with (25), then we find that $\sinh z = -i\sin(iz)$. After repeating this process for $\cos z$ and $\cosh z$ we obtain the following important relationships between the complex trigonometric and hyperbolic functions:

$$\sin z = -i\sinh(iz) \quad \text{and} \quad \cos z = \cosh(iz) \tag{27}$$

$\times(-i)$

$$\sinh z = -i\sin(iz) \quad \text{and} \quad \cosh z = \cos(iz). \tag{28}$$

Relations between the other trigonometric and hyperbolic functions can now be derived from (27) and (28). For example,

$$\tan(iz) = \frac{\sin(iz)}{\cos(iz)} = \frac{i\sinh z}{\cosh z} = i\tanh z.$$

We can also use (27) and (28) to derive hyperbolic identities from trigonometric identities. We next list some of the more commonly used hyperbolic identities. Each of the results in (29)–(32) is identical to its real analogue.

$$\sinh(-z) = -\sinh z \quad \cosh(-z) = \cosh z \tag{29}$$

$$\cosh^2 z - \sinh^2 z = 1 \tag{30}$$

$$\sinh(z_1 \pm z_2) = \sinh z_1 \cosh z_2 \pm \cosh z_1 \sinh z_2 \tag{31}$$

$$\cosh(z_1 \pm z_2) = \cosh z_1 \cosh z_2 \pm \sinh z_1 \sinh z_2 \tag{32}$$

EXAMPLE 4 A Hyperbolic Identity

Verify that $\cosh(z_1 + z_1) = \cosh z_1 \cosh z_2 + \sinh z_1 \sinh z_2$ for all complex z_1 and z_2.

Solution By (28), $\cosh(z_1 + z_2) = \cos(iz_1 + iz_2)$, and so by the trigonometric identity and additional applications of (27) and (28), we obtain:

$$\begin{aligned}
\cosh(z_1 + z_2) &= \cos(iz_1 + iz_2) \\
&= \cos iz_1 \cos iz_2 - \sin iz_1 \sin iz_2 \\
&= \cos iz_1 \cos iz_2 + (-i \sin iz_1)(-i \sin iz_2) \\
&= \cosh z_1 \cosh z_2 + \sinh z_1 \sinh z_2.
\end{aligned}$$

Inverse Trigonometric and Hyperbolic Functions

Inverse Sine

by using Definition to rewrite the equation $\sin w = z$

$$\frac{e^{iw}-e^{-iw}}{2i}=z \quad \text{or} \quad e^{2iw}-2ize^{iw}-1=0.$$

Because $e^{2iw}-2ize^{iw}-1=0$ is a quadratic equation in e^{iw} we can then use the quadratic formula to solve for e^{iw}:

$$e^{iw}=iz+\left(1-z^2\right)^{1/2}. \tag{1}$$

Since we are using the quadratic formula, we should keep in mind that the expression $\left(1-z^2\right)^{1/2}$ in (1) represents the *two* square roots of $1-z^2$. Finally, we solve for w using the complex logarithm:

$$iw=\ln\left[iz+\left(1-z^2\right)^{1/2}\right] \quad \text{or} \quad w=-i\ln\left[iz+\left(1-z^2\right)^{1/2}\right]. \tag{2}$$

Each value of w obtained from the second equation in (2) satisfies the equation $\sin w = z$. Therefore, we call the multiple-valued function defined by the second equation in (2) the inverse sine. We summarize this discussion in the following definition.

Definition [redacted] Inverse Sine

The multiple-valued function $\sin^{-1} z$ defined by:

$$\sin^{-1} z = -i \ln \left[iz + (1 - z^2)^{1/2} \right] \tag{3}$$

is called the **inverse sine.**

EXAMPLE 1 Values of Inverse Sine

Find all values of $\sin^{-1} \sqrt{5}$.

Solution By setting $z = \sqrt{5}$ in (3) we obtain:

$$\begin{aligned} \sin^{-1} \sqrt{5} &= -i \ln \left[i\sqrt{5} + \left(1 - \left(\sqrt{5}\right)^2\right)^{1/2} \right] \\ &= -i \ln \left[i\sqrt{5} + (-4)^{1/2} \right]. \end{aligned}$$

The two square roots $(-4)^{1/2}$ of -4 are found to be $\pm 2i$

$$\sin^{-1}\sqrt{5} = -i\ln\left[i\sqrt{5} \pm 2i\right] = -i\ln\left[\left(\sqrt{5} \pm 2\right)i\right].$$

Because $(\sqrt{5} \pm 2)\, i$ is a pure imaginary number with positive imaginary part (both $\sqrt{5}+2$ and $\sqrt{5}-2$ are positive), we have $|(\sqrt{5} \pm 2)\, i| = \sqrt{5} \pm 2$ and $\arg\left[(\sqrt{5} \pm 2)\, i\right] = \pi/2$. Thus, we have

$$\ln\left[\left(\sqrt{5} \pm 2\right) i\right] = \log_e\left(\sqrt{5} \pm 2\right) + i\left(\frac{\pi}{2} + 2n\pi\right)$$

for $n = 0, \pm 1, \pm 2, \ldots$. This expression can be simplified by observing that

$$\log_e\left(\sqrt{5} - 2\right) = \log_e \frac{1}{\sqrt{5}+2} = \log_e 1 - \log_e\left(\sqrt{5}+2\right) = 0 - \log_e\left(\sqrt{5}+2\right),$$

and so $\log_e(\sqrt{5} \pm 2) = \pm\log_e(\sqrt{5}+2)$. Therefore,

$$\begin{aligned} -i\ln\left[\left(\sqrt{5} \pm 2\right) i\right] &= -i\left[\log_e\left(\sqrt{5} \pm 2\right) + i\left(\frac{\pi}{2} + 2n\pi\right)\right] \\ &= -i\left[\pm\log_e\left(\sqrt{5}+2\right) + i\,\frac{(4n+1)\pi}{2}\right], \end{aligned}$$

and so

$$\sin^{-1}\sqrt{5} = \frac{(4n+1)\pi}{2} \pm i\log_e\left(\sqrt{5}+2\right)$$

for $n = 0, \pm 1, \pm 2, \ldots$.

to solve the equations $\cos w = z$ and $\tan w = z$.

Definition [redacted] Inverse Cosine and Inverse Tangent

The multiple-valued function $\cos^{-1} z$ defined by:

$$\cos^{-1} z = -i \ln\left[z + i\left(1 - z^2\right)^{1/2}\right] \tag{4}$$

is called the **inverse cosine**. The multiple-valued function $\tan^{-1} z$ defined by:

$$\tan^{-1} z = \frac{i}{2} \ln\left(\frac{i+z}{i-z}\right) \tag{5}$$

is called the **inverse tangent**.

Both the inverse cosine and inverse tangent are multiple-valued functions since they are defined in terms of the complex logarithm $\ln z$. As with the inverse sine, the expression $\left(1-z^2\right)^{1/2}$ in (4) represents the two square roots of the complex number $1-z^2$. Every value of $w = \cos^{-1} z$ satisfies the equation $\cos w = z$, and, similarly, every value of $w = \tan^{-1} z$ satisfies the equation $\tan w = z$.

Branches and Analyticity The inverse sine and inverse cosine are multiple-valued functions that can be made single-valued by specifying a single value of the square root to use for the expression $(1-z^2)^{1/2}$ and a single value of the complex logarithm to use in (3) or (4). The inverse tangent, on the other hand, can be made single-valued by just specifying a single value of $\ln z$ to use. For example, we can define a function f that gives a value of the inverse sine by using the principal square root and the principal value of the complex logarithm in (3). If, say, $z=\sqrt{5}$, then the principal square root of $1-\left(\sqrt{5}\right)^2=-4$ is $2i$, and

$$\text{Ln}\left(i\sqrt{5}+2i\right)=\log_e\left(\sqrt{5}+2\right)+\pi i/2.$$

Identifying these values in (3) gives:

$$f\left(\sqrt{5}\right)=\frac{\pi}{2}-i\log_e\left(\sqrt{5}+2\right)\approx 1.5708-1.4436i.$$

Thus, we see that the value of the function f at $z=\sqrt{5}$ is the value of $\sin^{-1}\sqrt{5}$ associated to $n=0$ and the square root $2i$ in Example 1.

A branch of a multiple-valued inverse trigonometric function may be obtained by choosing a branch of the square root function and a branch of the complex logarithm. Determining the domain of a branch defined in this manner can be quite involved. Because this is an elementary text, we will not discuss this topic further. On the other hand, the derivatives of branches of the multiple-valued inverse trigonometric functions are easily found using implicit differentiation. To see that this is so, suppose that f_1 is a branch of the multiple-valued function $F(z)=\sin^{-1}z$. If $w=f_1(z)$, then we know that $z=\sin w$. By differentiating both sides of this last equation with respect to z and applying the chain rule (6) ., we obtain:

$$1 = \cos w \cdot \frac{dw}{dz} \quad \text{or} \quad \frac{dw}{dz} = \frac{1}{\cos w}. \tag{6}$$

Now, from the trigonometric identity $\cos^2 w + \sin^2 w = 1$, we have $\cos w = \left(1 - \sin^2 w\right)^{1/2}$, and since $z = \sin w$, this may be written as $\cos w = \left(1 - z^2\right)^{1/2}$. Therefore, after substituting this expression for $\cos w$ in (6) we obtain the following result:

$$f_1'(z) = \frac{dw}{dz} = \frac{1}{(1 - z^2)^{1/2}}.$$

If we let $\sin^{-1} z$ denote the branch f_1, then this formula may be restated in a less cumbersome manner as:

$$\frac{d}{dz} \sin^{-1} z = \frac{1}{(1 - z^2)^{1/2}}.$$

Derivatives of Branches $\sin^{-1} z$, $\cos^{-1} z$, *and* $\tan^{-1} z$

$$\frac{d}{dz}\sin^{-1} z = \frac{1}{(1-z^2)^{1/2}} \tag{7}$$

$$\frac{d}{dz}\cos^{-1} z = \frac{-1}{(1-z^2)^{1/2}} \tag{8}$$

$$\frac{d}{dz}\tan^{-1} z = \frac{1}{1+z^2} \tag{9}$$

EXAMPLE 2 Derivative of a Branch of Inverse Sine

Let $\sin^{-1} z$ represent a branch of the inverse sine obtained by using the principal branches of the square root and the logarithm

Find the derivative of this branch at $z = i$.

Solution We note in passing that this branch is differentiable at $z = i$ because $1 - i^2 = 2$ is not on the branch cut of the principal branch of the square root function, and because $i\,(i) + \left(1 - i^2\right)^{1/2} = -1 + \sqrt{2}$ is not on the branch cut of the principal branch of the complex logarithm. Thus, by (7) we have:

$$\frac{d}{dz} \sin^{-1} z\Big|_{z=i} = \frac{1}{(1-z^2)^{1/2}}\Bigg|_{z=i} = \frac{1}{\left(1-i^2\right)^{1/2}} = \frac{1}{2^{1/2}}.$$

Using the principal branch of the square root, we obtain $2^{1/2} = \sqrt{2}$. Therefore, the derivative is $1/\sqrt{2}$ or $\frac{1}{2}\sqrt{2}$.

Definition [redacted] Inverse Hyperbolic Sine, Cosine, and Tangent

The multiple-valued functions $\sinh^{-1} z$, $\cosh^{-1} z$, and $\tanh^{-1} z$, defined by:

$$\sinh^{-1} z = \ln\left[z + \left(z^2+1\right)^{1/2}\right], \tag{10}$$

$$\cosh^{-1} z = \ln\left[z + \left(z^2-1\right)^{1/2}\right], \tag{11}$$

and

$$\tanh^{-1} z = \frac{1}{2}\ln\left(\frac{1+z}{1-z}\right) \tag{12}$$

are called the inverse hyperbolic sine, the inverse hyperbolic cosine, and the inverse hyperbolic tangent, respectively.

Derivatives of Branches $\sinh^{-1} z$, $\cosh^{-1} z$, *and* $\tanh^{-1} z$

$$\frac{d}{dz}\sinh^{-1} z = \frac{1}{\left(z^2+1\right)^{1/2}} \tag{13}$$

$$\frac{d}{dz}\cosh^{-1} z = \frac{1}{\left(z^2-1\right)^{1/2}} \tag{14}$$

$$\frac{d}{dz}\tanh^{-1} z = \frac{1}{1-z^2} \tag{15}$$

EXAMPLE 3 Inverse Hyperbolic Cosine

Let $\cosh^{-1} z$ represent the branch of the inverse hyperbolic cosine obtained by using the branch $f_2(z) = \sqrt{r}e^{i\theta/2}$, $0 < \theta < 2\pi$, of the square root and the principal branch of the complex logarithm. Find the following values.

(a) $\cosh^{-1} \dfrac{\sqrt{2}}{2}$ (b) $\left.\dfrac{d}{dz} \cosh^{-1} z\right|_{z=\sqrt{2}/2}$

Solution (a) In order to find $\cosh^{-1}\left(\frac{1}{2}\sqrt{2}\right)$, we use (11) with $z = \frac{1}{2}\sqrt{2}$ and the stated branches of the square root and logarithm. When $z = \frac{1}{2}\sqrt{2}$, we have that $z^2 - 1 = -\frac{1}{2}$. Since $-\frac{1}{2}$ has exponential form $\frac{1}{2}e^{i\pi}$, the square root given by the branch f_2 is:

$$f_2\left(\frac{1}{2}e^{i\pi}\right) = \sqrt{\frac{1}{2}}e^{i\pi/2} = \frac{1}{\sqrt{2}}i = \frac{\sqrt{2}}{2}i.$$

The value of our branch of the inverse cosine is then given by:

$$\cosh^{-1} \frac{\sqrt{2}}{2} = \ln\left[z + \left(z^2 - 1\right)^{1/2}\right] = \ln\left[\frac{\sqrt{2}}{2} + \frac{\sqrt{2}}{2}i\right],$$

where we take the value of the principal branch of the logarithm. Because $\left|\frac{1}{2}\sqrt{2} + \frac{1}{2}\sqrt{2}i\right| = 1$ and $\operatorname{Arg}\left(\frac{1}{2}\sqrt{2} + \frac{1}{2}\sqrt{2}i\right) = \frac{1}{4}\pi$, the principal branch of the logarithm is $\log_e 1 + i\left(\frac{1}{4}\pi\right) = \frac{1}{4}\pi i$. Therefore,

$$\cosh^{-1} \frac{\sqrt{2}}{2} = \frac{\pi}{4}i.$$

(b) From (14) we have:

$$\left.\frac{d}{dz}\cosh^{-1} z\right|_{z=\sqrt{2}/2} = \frac{1}{\left[\left(\sqrt{2}/2\right)^2 - 1\right]^{1/2}} = \frac{1}{(-1/2)^{1/2}}.$$

After using f_2 to find the square root in this expression we obtain:

$$\left.\frac{d}{dz}\cosh^{-1} z\right|_{z=\sqrt{2}/2} = \frac{1}{\sqrt{2}i/2} = -\sqrt{2}i.$$

Each of the infinitely many functions in is called a **branch** of the logarithm. The negative real axis is known as a **branch cut** and
The branch for $n = 0$ is called the **principal branch** of $\ln z$.

$$\ln z = \operatorname{Ln} z \pm 2n\pi i \qquad (n = 1, 2, \cdots).$$

Calculate $\sin\left(\frac{\pi}{4}+i\right)$.

Solution.

$$
\begin{aligned}
\sin\left(\frac{\pi}{4}+i\right) &= \frac{1}{2i}(e^{i(\pi/4+i)} - e^{-i(\pi/4+i)}) \\
&= \frac{1}{2i}(e^{-1}e^{\pi i/4} - ee^{-\pi i/4}) \\
&= \frac{1}{2i}\left(e^{-1}(\cos\frac{\pi}{4} + i\sin\frac{\pi}{4}) - e(\cos\frac{\pi}{4} - i\sin\frac{\pi}{4})\right) \\
&= \frac{\sqrt{2}}{4}\left(e+\frac{1}{e}\right) + \frac{\sqrt{2}}{4}\left(e-\frac{1}{e}\right)i
\end{aligned}
$$

Compute $\cos\left(\frac{\pi}{3}+i\right)$.

Solution.

$$\begin{aligned}
\cos\left(\frac{\pi}{3}+i\right) &= \frac{1}{2}(e^{i(\pi/3+i)}+e^{-i(\pi/3+i)}) \\
&= \frac{1}{2}(e^{-1}e^{\pi i/3}+ee^{-\pi i/3}) \\
&= \frac{1}{2}\left(e^{-1}(\cos\frac{\pi}{3}+i\sin\frac{\pi}{3})+e(\cos\frac{\pi}{3}-i\sin\frac{\pi}{3})\right) \\
&= \frac{1}{4}\left(e+\frac{1}{e}\right)-\frac{\sqrt{3}i}{4}\left(e-\frac{1}{e}\right)
\end{aligned}$$

For $z \in \mathbb{C}$, show that:
(a) $\sin \overline{z} = \overline{\sin z}$;
(b) $\cosh \overline{z} = \overline{\cosh z}$.

Proof. (a) Using the definition of $\sin z$, we have

$$\begin{aligned}\sin \overline{z} &= \frac{e^{i\overline{z}} - e^{-i\overline{z}}}{2i} = \frac{e^{\overline{-iz}} - e^{\overline{iz}}}{\overline{-2i}} = \overline{\left(\frac{e^{-iz} - e^{iz}}{-2i}\right)} = \overline{\left(\frac{e^{iz} - e^{-iz}}{2i}\right)} \\ &= \overline{\sin z}\end{aligned}$$

(b) Recall that for $z = x + iy$, we have that

$$e^{\overline{z}} = e^{x-iy} = e^x e^{-iy} = \overline{e^x e^{iy}} = \overline{e^z}.$$

Thus

$$\cosh \overline{z} = \frac{e^{\overline{z}} - e^{-\overline{z}}}{2} = \overline{\left(\frac{e^z - e^{-z}}{2}\right)} = \overline{\cosh z}$$

Show that

$$\tanh^{-1} z = \frac{1}{2}\log\left(\frac{1+z}{1-z}\right).$$

Solution. If $w = \tanh^{-1} z$, then

$$z = \tanh w \quad = \quad \frac{\sinh w}{\cosh w} = \frac{e^w - e^{-w}}{e^w + e^{-w}} = \frac{e^{2w} - 1}{e^{2w} + 1}$$

Thus

$$\begin{aligned}
z(e^{2w} + 1) &= e^{2w} - 1 \\
e^{2w} &= \frac{z+1}{1-z} \\
2w \log e &= \log\left(\frac{z+1}{1-z}\right) \\
w &= \frac{1}{2}\log\left(\frac{z+1}{1-z}\right)
\end{aligned}$$

Hence

$$\tanh^{-1} z = \frac{1}{2}\log\left(\frac{z+1}{1-z}\right).$$

Find all solutions of the equation $\tanh z = i$ and express them in the form $x+iy$.

Solution.

Applying the inverse hyperbolic function in both sides we have

$$z = \tanh^{-1}(i).$$

Using the formula , we have

$$\begin{aligned}\tanh^{-1}(i) &= \frac{1}{2}\log\left(\frac{1+i}{1-i}\right)\\ &= \frac{1}{2}\log(i)\end{aligned}$$

Using the definition of log for i with $r=1$ and $\Theta = \pi/2$ we have that

$$\log(i) = \ln 1 + i\left(\frac{\pi}{2}+2n\pi\right) = i\left(\frac{\pi}{2}+2n\pi\right) \quad (n \in \mathbb{Z}).$$

Therefore

$$z = \tanh^{-1}(i) = \frac{1}{2}\left[i\left(\frac{\pi}{2}+2n\pi\right)\right] = i\left(\frac{\pi}{4}+n\pi\right)$$

for $n \in \mathbb{Z}$.

Determine Whether the following functions are harmoniz.

1 a) $f(x,y) = \operatorname{Im}(\cos z)$ $\qquad \cos z = \dfrac{e^{iz} + e^{-iz}}{2}$ $\qquad z = x+iy$

$$\Rightarrow \cos z = \frac{1}{2}\left(e^{i(x+iy)} + e^{-i(x+iy)}\right) = \frac{1}{2}\left(e^{ix}e^{-y} + e^{-ix}e^{y}\right)$$

$$= \frac{1}{2}\left[(\cos x + i\sin x)e^{-y} + (\cos x - i\sin x)e^{y}\right]$$

$$= \cos x \,\frac{e^{y}+e^{-y}}{2} + i\sin x \,\frac{e^{-y}-e^{y}}{2} = \cos x \cosh y - i\sin x \sinh y$$

$$f(x,y) = \operatorname{Im}(\cos z) = -\sin x \sinh y$$

$f_x = -\cos x \sinh y \qquad f_{xx} = \sin x \sinh y$

$f_y = -\sin x \cosh y \qquad f_{yy} = -\sin x \sinh y$ $\qquad f_{xx} + f_{yy} = 0$

$\Rightarrow f(x,y)$ harmonik.

$\Rightarrow f(x,y)$ harmonik.

b) $g(x,y) = Re(\sin z)$ $\quad \sin z = \dfrac{e^{iz} - e^{-iz}}{2i}$ $\quad z = x + iy$

$$\sin z = \frac{1}{2i}\left(e^{i(x+iy)} - e^{-i(x+iy)}\right) = -\frac{i}{2}\left(e^{ix}e^{-y} - e^{-ix}e^{y}\right)$$

$$= -\frac{i}{2}\left[(\cos x + i\sin x)e^{-y} - (\cos x - i\sin x)e^{y}\right]$$

$$= \frac{1}{2}\left[(-i\cos x + \sin x)e^{-y} + (i\cos x + \sin x)e^{y}\right]$$

$$= \sin x \frac{e^{-y} + e^{y}}{2} + i\cos x \frac{e^{y} - e^{-y}}{2} = \sin x \cosh y + i\cos x \sinh y$$

$g(x,y) = Re(\sin z) = \sin x \cosh y$

$$\left.\begin{array}{ll} g_x = \cos x \cosh y & g_{xx} = -\sin x \cosh y \\ g_y = \sin x \sinh y & g_{yy} = \sin x \cosh y \end{array}\right\} g_{xx} + g_{yy} = 0$$

$\Rightarrow g(x,y)$ harmonik

$$\cos z = \frac{e^{it}+e^{-it}}{2} = 3$$

$$e^{2iz} - 6e^{iz} + 1 = 0 \qquad e^{iz} = t \Rightarrow t^2 - 6t + 1 = 0$$

$$t_{1,2} = 3 \pm \sqrt{9-1} = 3 \pm 2\sqrt{2}$$

$$e^{iz} = 3 + 2\sqrt{2} \Rightarrow iz = \ln(3+2\sqrt{2}) \qquad |3+2\sqrt{2}| = 3+2\sqrt{2}$$

$$\arg(3+2\sqrt{2}) = 0 + 2\pi n$$

$$\Rightarrow iz = \log_e(3+2\sqrt{2}) + 2\pi n i$$

$$\Rightarrow z_1 = 2\pi n - i\log_e(3+2\sqrt{2}) \quad n = 0, \pm 1, \ldots$$

$$e^{it} = 3 - 2\sqrt{2} \Rightarrow it = \log_e(3-2\sqrt{2}) + 2\pi n \hat{i}$$

$$\Rightarrow z_2 = 2\pi n - i\log_e(3-2\sqrt{2}) \quad n = 0, \pm 1, \ldots$$

$$\Rightarrow \boxed{z = 2\pi n \pm i\log_e(3 \mp 2\sqrt{2}) \quad n = 0, \pm 1, \ldots}$$

$$\left(\log_e(3-2\sqrt{2}) = -\log_e(3+2\sqrt{2})\right)$$

$$(\sqrt{3}-i)^{i} = e^{i\ln(\sqrt{3}-i)}$$

$$|\sqrt{3}-i| = 2$$

$$\mathrm{Arg}(\sqrt{3}-i) = -\frac{\pi}{6}$$

$$\Rightarrow \log(\sqrt{3}-i) = \log_e 2 + i\left(-\frac{\pi}{6}\right)$$

$$(\sqrt{3}-i)^{i} = e^{i\left(\log_e 2 + i\frac{\pi}{6}\right)} = e^{\frac{\pi}{6}}\, e^{i\log_e 2} = e^{\pi/6}\left(\cos(\log_e 2) + i\sin(\log_e 2)\right)$$